

Zero Order Accelerated Gradient Descent By Linear Mixing

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Assumption 1 *We assume that the smoothness constant of f is $L \geq 1$. Otherwise, we can replace it with 1. Also assume that $0 < \delta < 1/e$.*

We use the linear mixing framework to design an accelerated zero-order gradient descent algorithm. The idea is to query gradient descent y_t and mirror descent z_t at the same point x_t , and then linearly mix the answers $x_{t+1} = \lambda z_t + (1 - \lambda)y_t$. The algorithm can be described as

$$\begin{aligned} x_{t+1} &= \lambda y_t + (1 - \lambda)z_t \\ y_{t+1} &= x_{t+1} - \eta \tilde{\nabla} f(x_{t+1}) \\ z_{t+1} &= \text{Mirr}_{z_t}(\alpha \tilde{\nabla} f(x_{t+1})) \end{aligned}$$

The first thing to do is to deal with quadratic error.

Lemma 1 (Quadratic Error)

$$\|\tilde{\nabla} f(x) - \frac{1}{r} \sum_{i=1}^r p_i p_i^\top \nabla f(x)\| \leq L\sigma \frac{1}{r} \sum_{i=1}^r \|p_i\|^3$$

Proof We have

$$\begin{aligned} \frac{f(x + \sigma p) - f(x - \sigma p)}{2\sigma} p - p p^\top \nabla f(x) &= \frac{1}{2\sigma} \left[\int_{-1}^1 \frac{d}{dt} f(x + t\sigma p) - \sigma p p^\top \nabla f(x) dt \right] \\ &= \frac{1}{2\sigma} \left[\int_{-1}^1 \langle \nabla f(x + t\sigma p) - \nabla f(x), \sigma p \rangle p dt \right] \end{aligned}$$

Now, by L -smoothness,

$$\left| \int_{-1}^1 \langle \nabla f(x + t\sigma p) - \nabla f(x), \sigma p \rangle p dt \right| \leq \int_{-1}^1 L\sigma^2 |t| \|p\|^3 dt$$

Therefore,

$$\left| \frac{f(x + \sigma p) - f(x - \sigma p)}{2\sigma} p - p p^\top \nabla f(x) \right| \leq L\sigma \|p\|^3$$

Then by triangle inequality,

$$\|\tilde{\nabla} f(x) - \frac{1}{r} \sum_{i=1}^r p_i p_i^\top \nabla f(x)\| \leq L\sigma \frac{1}{r} \sum_{i=1}^r \|p_i\|^3.$$

■

Unfortunately, this bound is necessary. Ideally we would like to just say that $f(x + \sigma p) - f(x - \sigma p)/2\sigma p = pp^\top \nabla g(x)$ for some function $g(x)$ related to $f(x)$. Possibly a smoothed variant f_σ . But this cannot possibly hold for every p unless f is quadratic in a neighborhood of x . To see this, we can use MVT to expand the LHS as $\langle \nabla f(x + \theta \sigma p), p \rangle p$. By taking $p = e_i$, we see that $e_i^\top \nabla g(x) = e_i^\top \nabla f(x + \theta_i \sigma e_i)$. Sending $\sigma \rightarrow 0$, $\nabla f(x + \theta_i \sigma e_i) \rightarrow \nabla f(x)$ under L -smoothness assumption, meaning that $\nabla g(x) = \nabla f(x)$. This would say that f is quadratic, which is not true in general.

Recall the following lemma from before.

Lemma 2 *Let X follow χ_d^2 distribution and $0 < \delta < 1/e$. Then there exists L_δ such that for every d , with*

$$A := \{dL_\delta^2 \leq X \leq 3d \log(C/\delta)\}$$

we have

$$\mathbb{E}[(X - d); A] = 0, \quad \mathbb{P}(A) \geq 1 - \delta$$

for some absolute constant C .

These lemmas tell us how much progress the gradient descent step is making.

Lemma 3 (? Corollary 7.3.2) *Let $P \in \mathbb{R}^{d \times r}$ be a matrix with i.i.d. Gaussian entries. Define $\|A\|_{\text{op}} = \sup_{u \in \mathbb{S}^{r-1}} \|Au\|_2$. Then*

$$\mathbb{P}(\|P\|_{\text{op}} \geq \sqrt{d} + \sqrt{r} + s) \leq 2e^{-cs^2}$$

Proposition 4 *Define $\chi = \log(CrT/\delta)$ and let $P_t = [p_{t,1} \ \cdots \ p_{t,r}]$. Define the bounded operator norm event*

$$\mathfrak{D}_t = \left\{ \frac{1}{\sqrt{r}} \|P\|_{\text{op}} \leq C \frac{d}{r} \sqrt{\log(cTr/\delta)} \right\}$$

Then on $\mathfrak{D}_t$, for every $0 \leq t \leq T-1$,

$$\left\| \frac{1}{r} \sum_{i=1}^r p_{t,i} p_{t,i}^\top \nabla f(x_t) \right\|^2 \leq \frac{Cd}{r} \log(cTr/\delta) \cdot \frac{1}{r} \sum_{i=1}^r |p_{t,i}^\top \nabla f(x_t)|^2$$

Furthermore,

$$\mathbb{P}\left(\bigcap_{t=0}^{T-1} \mathfrak{D}_t\right) \geq 1 - \delta/3$$

Proof Write $P = [p_{t,1} \ \cdots \ p_{t,r}] \in \mathbb{R}^{d \times r}$. By choosing $s = \sqrt{1/c \log(2/\delta)}$ above, we know that

$$\frac{1}{\sqrt{r}} \|P\|_{\text{op}} \leq 1 + \sqrt{\frac{d}{r}} + \frac{1}{c\sqrt{r}} \sqrt{\log(c/\delta)} \leq C \sqrt{\frac{d}{r}} \sqrt{\log(c/\delta)}$$

with probability at least $1 - \delta$. As $PP^\top = \sum_{i=1}^r p_{t,i} p_{t,i}^\top$, we see that

$$\left\| \frac{1}{r} PP^\top \nabla f(x_t) \right\|^2 \leq \left(\frac{1}{\sqrt{r}} \|P\|_{\text{op}} \right)^2 \frac{1}{r} \|P^\top \nabla f(x_t)\|^2 \leq \frac{Cd}{r} \log(c/\delta) \frac{1}{r} \sum_{i=1}^r |p_{t,i}^\top \nabla f(x_t)|^2$$

We then rescale δ to δ/Tr to complete the proof. ■

Lemma 5 *Define the bounded ps event*

$$\mathfrak{B}_{t+1} := \bigcap_{i=1}^r \left\{ \sqrt{dL_{\delta/rT}} \leq \|p_{t+1,i}\| \leq \sqrt{3d\chi} \right\}$$

Then on $\mathfrak{B}_{t+1}$, as long as $\eta \leq r/(12Ld\chi)$, we have

$$f(y_{t+1}) \leq f(x_{t+1}) - \frac{3\eta}{4r} \sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 + \frac{\eta^2}{\alpha} \|\nabla f(x_{t+1})\|^2 + (\alpha + 1)CL^2\sigma^2\chi^3d^3$$

Furthermore,

$$\mathbb{P} \left(\bigcap_{t=0}^{T-1} \mathfrak{B}_t \right) \geq 1 - \delta/3$$

Proof By L -smoothness,

$$\begin{aligned} f(y_{t+1}) &\leq f(x_{t+1}) + \langle \nabla f(x_{t+1}), y_{t+1} - x_{t+1} \rangle + \frac{L}{2} \|y_{t+1} - x_{t+1}\|^2 \\ &= f(x_{t+1}) - \eta \langle \nabla f(x_{t+1}), \tilde{\nabla} f(x_{t+1}) \rangle + \frac{L\eta^2}{2} \|\tilde{\nabla} f(x_{t+1})\|^2 \end{aligned}$$

Writing $\tilde{\nabla} f(x_{t+1}) = \frac{1}{r} \sum_{i=1}^r p_i p_i^\top \nabla f(x_{t+1}) + R_{t+1}$, we get

$$\begin{aligned} f(y_{t+1}) &\leq f(x_{t+1}) - \frac{\eta}{r} \sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 - \eta \langle \nabla f(x_{t+1}), R_{t+1} \rangle \\ &\quad + L\eta^2 \left\| \frac{1}{r} \sum_{i=1}^r p_i p_i^\top \nabla f(x_{t+1}) \right\|^2 + L\eta^2 \|R_{t+1}\|^2 \end{aligned}$$

Now, $2\langle \eta \nabla f(x_{t+1}), R_{t+1} \rangle \leq \eta^2/\alpha \|\nabla f(x_{t+1})\|^2 + \alpha R_{t+1}^2$ by Cauchy-Schwarz and Young's inequality.

Define the event where p_i are bounded:

$$\mathfrak{B}_{t+1} := \bigcap_{i=1}^r \left\{ \sqrt{dL_{\delta/rT}} \leq \|p_{t+1,i}\| \leq \sqrt{3d\chi} \right\}$$

We know that $\mathbb{P}(\mathfrak{B}_{t+1}) \geq 1 - \delta/rT$. Thus,

$$\begin{aligned} f(y_{t+1}) &\leq f(x_{t+1}) - \frac{\eta}{r} \sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 + \frac{\eta^2}{\alpha} \|\nabla f(x_{t+1})\|^2 + (L\eta^2 + \alpha) \|R_{t+1}\|^2 \\ &\quad + \frac{CLd\eta^2 \log(Tr/\delta)}{r} \cdot \frac{1}{r} \sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 \end{aligned}$$

From quadratic error lemma we know that $|R_{t+1}| \leq 3L\sigma \log^{3/2}(1/\delta)d^{3/2}$. Now by choosing $\eta \leq r/(4CLd\chi)$, we know that $-\eta + CLd\chi\eta^2/r \leq -3/4\eta$. Therefore our final bound becomes

$$f(y_{t+1}) \leq f(x_{t+1}) - \frac{3\eta}{4r} \sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 + \frac{\eta^2}{\alpha} \|\nabla f(x_{t+1})\|^2 + (\alpha + 1)CL^2\sigma^2\chi^3d^3$$

for some absolute constant C . We used that $L\eta^2 < 1$ and $0 < \delta < 1/e$. ■

We can copy the mirror descent lemma from last time:

Lemma 6 *If $z_{t+1} = \text{Mirr}_{z_t}(\alpha \tilde{\nabla} f(x_{t+1}))$, then*

$$\forall u \in \mathbb{R}^d \quad \alpha \langle \tilde{\nabla} f(x_{t+1}), z_t - u \rangle \leq \frac{\alpha^2}{2} \|\tilde{\nabla} f(x_{t+1})\|^2 + V_{z_t}(u) - V_{z_{t+1}}(u).$$

The standard coupling lemma comes from using $\|\nabla f(x_{t+1})\|^2 \leq 2L(f(x_{t+1}) - f(y_{t+1}))$. However this doesn't work here, as we are using an estimator. But we can fix this using the following lemma.

Lemma 7 *On $\mathfrak{B}_{t+1}$, we have*

$$\|\tilde{\nabla} f(x_{t+1})\|^2 \leq \frac{Cd\chi}{r\eta} [f(x_{t+1}) - f(y_{t+1}) + \eta^2 \|\nabla f(x_{t+1})\|^2 + C'L^2\sigma^2\chi^3d^3]$$

Proof As before we have

$$\begin{aligned} \|\tilde{\nabla} f(x_{t+1})\|^2 &\leq 2 \left\| \frac{1}{r} \sum_{i=1}^r p_i p_i^\top \nabla f(x_{t+1}) \right\|^2 + 2 \|R_{t+1}\|^2 \\ &\leq 2 \frac{Cd}{r} \log(cTr/\delta) \cdot \frac{1}{r} \sum_{i=1}^r |p_{t,i}^\top \nabla f(x_t)|^2 + 2 \|R_{t+1}\|^2 \\ &\leq \frac{Cd}{r} \log(cTr/\delta) \cdot \frac{1}{r} \sum_{i=1}^r |p_{t,i}^\top \nabla f(x_t)|^2 + C'L^2\sigma^2 \log^3(rT/\delta) d^3 \end{aligned}$$

From before we know that

$$\frac{3\eta}{4r} \sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 \leq f(x_{t+1}) - f(y_{t+1}) + \eta^2 \|\nabla f(x_{t+1})\|^2 + C'L^2\sigma^2 \log^3(rT/\delta) d^3$$

Therefore an upper bound on $\|\tilde{\nabla} f(x_{t+1})\|^2$ is

$$\frac{Cd\chi}{r\eta} [f(x_{t+1}) - f(y_{t+1}) + \eta^2 \|\nabla f(x_{t+1})\|^2 + C'L^2\sigma^2 \log^3(rT/\delta) d^3] + C'L^2\sigma^2\chi^3d^3$$

As $\eta < 1$ we can absorb the last term into the first by changing the constant C' which gives the statement of the lemma. $\blacksquare$

This lemma is the same as before.

Lemma 8 When λ is chosen so $\frac{1-\lambda}{\lambda} = \frac{4d\chi\alpha}{\eta}$, the “regret” is bounded by

$$\alpha\langle \nabla f(x_{t+1}), x_{t+1} - u \rangle - \alpha\langle \nabla f(x_{t+1}), z_t - u \rangle \leq \frac{4d\chi\alpha^2}{\eta}(f(y_t) - f(x_{t+1}))$$

Proof We have by convexity and the definition of $x_{t+1} = \lambda z_t + (1-\lambda)y_t$,

$$\begin{aligned} & \alpha\langle \nabla f(x_{t+1}), x_{t+1} - u \rangle - \alpha\langle \nabla f(x_{t+1}), z_t - u \rangle \\ &= \alpha\langle \nabla f(x_{t+1}), x_{t+1} - z_t \rangle = \frac{(1-\lambda)\alpha}{\lambda}\langle \nabla f(x_{t+1}), y_t - x_{t+1} \rangle \leq \frac{(1-\lambda)\alpha}{\lambda}(f(y_t) - f(x_{t+1})) \end{aligned}$$

Now let $\lambda \in (0, 1)$ satisfy $\frac{1-\lambda}{\lambda} = \frac{4d\chi\alpha}{\eta}$ and we are done. $\blacksquare$

Now that this is done, we move onto the part of the proof which is not straightforward. The main idea from here is that, the Martingale term can be bounded by the “Energy” $V_{z_t}(x^*)$, which itself can be bounded by the sum of the gradient norms $\sum \|\nabla f(x_t)\|^2$ and $\Theta \geq V_{z_0}(x^*)$.

This highlights a very general principle. To deal with martingale error terms with high probability, try bounding them by some other part of the equation which should have typical size $\Theta(1)$. In this case, that will be $V_{z_t}(x^*)$ for us. However the route to doing this is usually not straightforward.

This lemma reveals the martingale term we need to bound.

Lemma 9

$$\begin{aligned} & \alpha\langle \nabla f(x_{t+1}), z_t - u \rangle \\ & \leq \alpha\langle \tilde{\nabla} f(x_{t+1}), z_t - u \rangle + \alpha \sum_{i=1}^r \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle + C' T \alpha^2 \|R_{t+1}\|^2 + \frac{1}{4T} V_{z_t}(u). \end{aligned}$$

Proof

$$\langle \nabla f(x_{t+1}), z_t - u \rangle = \langle \tilde{\nabla} f(x_{t+1}), z_t - u \rangle + \sum_{i=1}^r \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle + \langle R_{t+1}, z_t - u \rangle$$

Now, by Young’s inequality, $\alpha\langle R_{t+1}, z_t - u \rangle \leq C' \alpha^2 T \|R_{t+1}\|^2 + \frac{1}{8T} \|z_t - u\|^2$, by strong convexity of Bregman divergence, $V_{z_t}(u) \geq \frac{1}{2} \|z_t - u\|^2$, so we arrive at the statement of the lemma. $\blacksquare$

Notice that $\sum_{t=0}^{T-1} \sum_{i=1}^r \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle$ is a martingale. This is the key idea behind why the rest of this proof works.

Definition 10 (Martingale Difference Sequence) $\{\xi_t\}_{t \geq 1}$ is a martingale difference sequence wrt filtration $\mathcal{F}_t$ if $\xi_t \in \mathcal{F}_t$ and

$$\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$$

The main intuition is that in a martingale difference sequence, the variance of the full sum is the sum of the variance of each ξ_t , which is vastly superior than including all the cross covariance terms $\text{Cov}(\xi_i, \xi_j)$. This improves our bound substantially.

Lemma 11 Define $S_T = \sum_{t=1}^T \xi_t$. Then

$$\mathbb{E}[S_T^2] = \sum_{t=1}^T \mathbb{E}[\xi_t^2]$$

Proof We have

$$\mathbb{E}[M_T^2] = \sum \mathbb{E}[\xi_t^2] + 2\mathbb{E}\left[\sum_{i<j} \xi_i \xi_j\right]$$

Now,

$$\mathbb{E}\left[\sum_{i<j} \xi_i \xi_j\right] = \sum_{i<j} \mathbb{E}[\mathbb{E}[\xi_i \xi_j \mid \mathcal{F}_i]] = \sum_{i<j} \mathbb{E}[\xi_i \mathbb{E}[\xi_j \mid \mathcal{F}_i]] = 0.$$

■

That is the heart of why the rest of this proof should work at all.

Lemma 12 Define

$$\mathfrak{M}_{t+1} = \left\{ \left| \sum_{t=0}^{\tau-1} \sum_{i=1}^r \alpha \langle (I - p_i p_i^\top) \nabla f(x_{t+1}, z_t - u) \rangle \right| \leq \alpha \sqrt{cr \sum_{t=0}^{\tau-1} \chi^{3/2} \|\nabla f(x_t)\|^2 V_{z_t}(u)} \right\}$$

Then

$$\mathbb{P}\left(\bigcap_{t=0}^{T-1} \mathfrak{M}_t\right) \geq 1 - \delta/3.$$

Proof From Lemma ?? above, we know that

$$\begin{aligned} & \left| \sum_{t=0}^{\tau-1} \sum_{i=1}^r \alpha \langle (I - p_i p_i^\top) \nabla f(x_{t+1}, z_t - u) \rangle \right| \\ & \leq c\theta \sum_{t=0}^{\tau-1} \sum_{i=1}^r \log(crT/\delta) \|\nabla f(x_{t+1})\|^2 \|z_t - u\|^2 + \frac{1}{\theta} \log\left(\frac{crT}{\delta}\right) \end{aligned}$$

Using $V_{z_t}(u) \geq \frac{1}{2} \|z_t - u\|^2$ and balancing the two terms by carefully choosing θ we arrive at the statement of the lemma. ■

The rest of this proof is on bounding $V_{z_t}(x^*)$.

Lemma 13 (Descent) *On $\bigcap (B_t \cap \mathfrak{D}_t \cap \mathfrak{M}_t)$, for all $1 \leq \tau \leq T$, we have*

$$\begin{aligned}
& \sum_{t=0}^{\tau-1} \alpha \langle \nabla f(x_t), x_t - x^* \rangle \\
& \leq \frac{CLd^2\chi^2}{r^2} \alpha^2 (f(y_0) - f(y_\tau)) + C\alpha T^2 L^2 \sigma^2 \chi^3 d^3 + \frac{\alpha}{T} \sum_{t=0}^{\tau-1} V_{z_t}(x^*) \\
& + \alpha \sqrt{\frac{c}{r} \sum_{t=0}^{\tau-1} \chi^{3/2} \|\nabla f(x_t)\|^2 V_{z_t}(x^*)} + \sum_{t=0}^{\tau-1} \frac{\alpha^2}{3L} \|\nabla f(x_{t+1})\|^2 + CTL^3 \sigma^2 \chi^5 d^5 \alpha^2 \\
& + V_{z_0}(x^*) - V_{z_\tau}(x^*)
\end{aligned}$$

Proof By Lemma 8, we know that

$$\sum_{t=0}^{\tau-1} \alpha \langle \nabla f(x_t), x_t - x^* \rangle \leq \sum_{t=0}^{\tau-1} \frac{4d\chi\alpha^2}{\eta} (f(y_t) - f(x_{t+1})) + \sum_{t=0}^{\tau-1} \alpha \langle \nabla f(x_{t+1}), z_t - u \rangle$$

By Lemma 9, we know that

$$\begin{aligned}
& \sum_{t=0}^{\tau-1} \alpha \langle \nabla f(x_{t+1}), z_t - u \rangle \\
& \leq \alpha \sum_{t=0}^{\tau-1} \left[\langle \tilde{\nabla} f(x_{t+1}), z_t - u \rangle + \frac{1}{r} \sum_{i=1}^r \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle + 2T \|R_{t+1}\|^2 + \frac{1}{T} V_{z_t}(u) \right]
\end{aligned}$$

From above we know that

$$\alpha \sum_{t=0}^{\tau-1} \sum_{i=1}^r \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle \leq \alpha \sqrt{cr \sum_{t=0}^{\tau-1} \chi^{3/2} \|\nabla f(x_t)\|^2 V_{z_t}(u)}$$

We then know that

$$\begin{aligned}
& \sum_{t=0}^{\tau-1} \alpha \langle \tilde{\nabla} f(x_{t+1}), z_t - u \rangle \\
& \leq \sum_{t=0}^{\tau-1} \frac{Cd\chi\alpha^2}{r\eta} [f(x_{t+1}) - f(y_{t+1}) + \frac{\eta^2}{\alpha} \|\nabla f(x_{t+1})\|^2 + C'(\alpha + 1)L^2 \sigma^2 \chi^3 d^3]
\end{aligned}$$

Recall that $\|R_{t+1}\|^2 \leq C'L^2\sigma^2\chi^3d^3$ and now choose $\eta = r/(4CLd\chi)$. Putting all of it together, we get:

$$\begin{aligned}
& \sum_{t=0}^{\tau-1} \alpha \langle \nabla f(x_t), x_t - x^* \rangle \\
& \leq \frac{CLd^2\chi^2}{r^2} \alpha^2 (f(y_0) - f(y_\tau)) + C'\alpha^2 T^2 L^2 \sigma^2 \chi^3 d^3 + \frac{1}{4T} \sum_{t=0}^{\tau-1} V_{z_t}(x^*) \\
& + \alpha \sqrt{\frac{c}{r} \sum_{t=0}^{\tau-1} \chi^{3/2} \|\nabla f(x_t)\|^2 V_{z_t}(x^*)} + \sum_{t=0}^{\tau-1} \frac{\alpha}{3L} \|\nabla f(x_{t+1})\|^2 + C'(\alpha^3 + \alpha^2) TL^3 \sigma^2 \chi^5 d^5 \\
& + V_{z_0}(x^*) - V_{z_\tau}(x^*)
\end{aligned}$$

■